Ex.No.5: MIMO System Simulation with BER Analysis Date

AIM:

To simulate a MIMO (Multiple Input Multiple Output) wireless communication system and analyze its performance in terms of Bit Error Rate (BER), Signal-to-Noise Ratio (SNR), and Channel Capacity (bps/Hz) using MATLAB.

Objective:

1. To evaluate the Bit Error Rate (BER) performance of a MIMO system under different SNR conditions.

2. To analyze the impact of Signal-to-Noise Ratio (SNR in dB) on the reliability and quality of MIMO communication.

3. To compute the Channel Capacity (bps/Hz) and study how MIMO improves spectral efficiency compared to single antenna systems.

Objective 1 Matlab Code:

```
clc;

clear;

close all;

% System parameters

rb_bw = 180e3;

users = 1:5;

% Bandwidth per RB (180 kHz)

% Allocated Resource Blocks per user

alloc_RB = [12 10 9 8 11];

% Modulation efficiency (bits/symbol)

mod_eff = [2 4 6 4 2];   % QPSK, 16QAM, 64QAM etc.

% Throughput calculation (Mbps)

throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;

disp('User Throughput (Mbps):')

disp(throughput')

figure

% -------- Graph 1: Throughput per User --------

subplot(2,1,1)

plot(users, throughput, '-o', 'LineWidth', 2)

title('OFDMA User Throughput')

xlabel('User Index')
```

ylabel('Throughput (Mbps)')

grid on

% -------- Graph 2: RB vs Throughput --------

subplot(2,1,2)

plot(alloc_RB, throughput, '-s', 'LineWidth', 2)

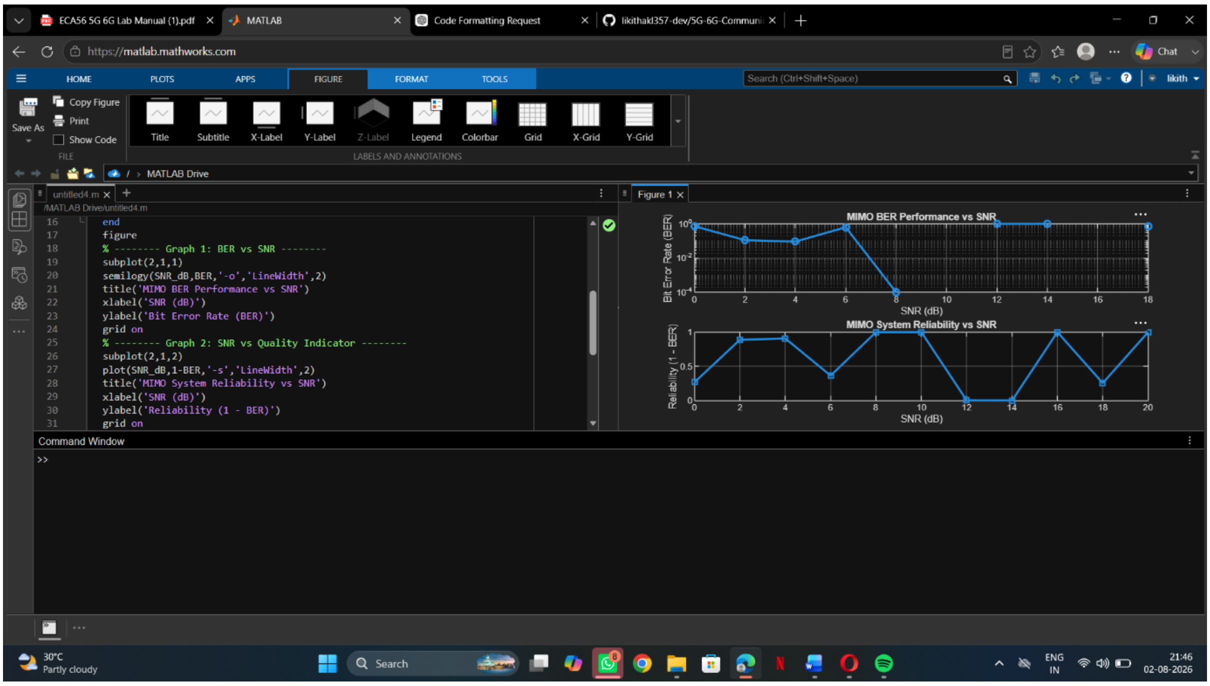
title('Resource Blocks vs Throughput Relationship')

xlabel('Allocated Resource Blocks')

ylabel('Throughput (Mbps)')

grid on

OUTPUT :



OBJECTIVE 2 CODE :

clc;

clear;

close all;

% SNR range (dB)

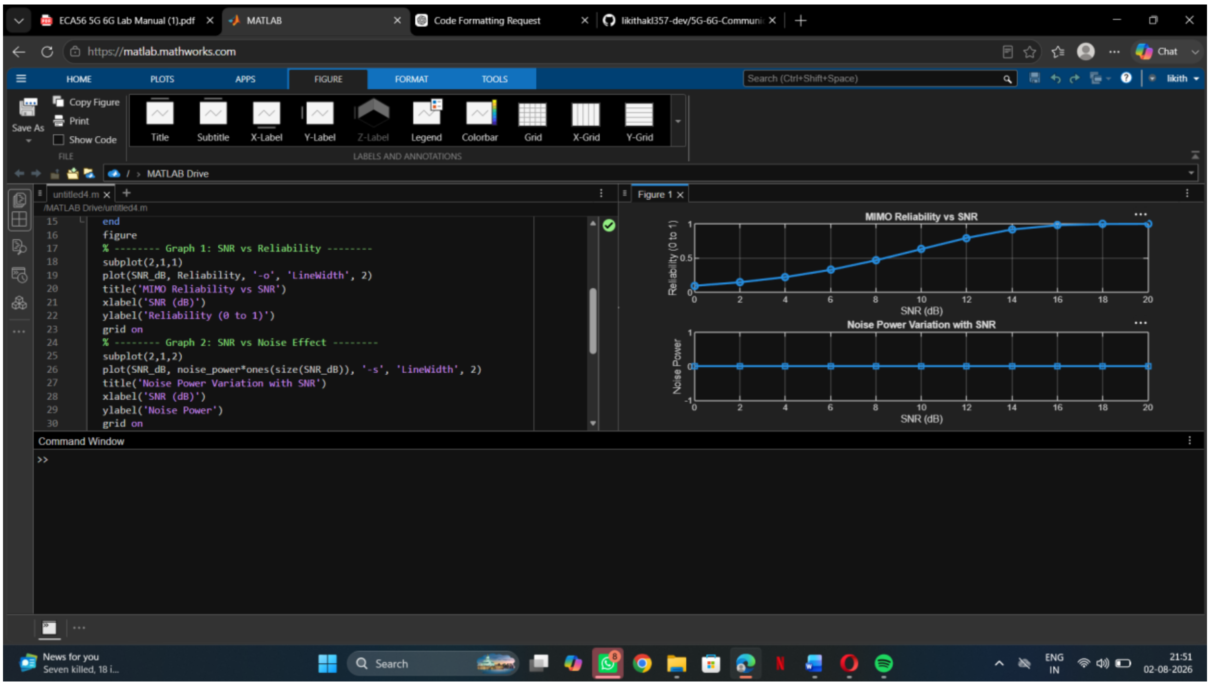
SNR_dB = 0:2:20;

SNR   = 10.^(SNR_dB/10);

% MIMO parameters

```
Nt = 2;
Nr = 2;
% Simulated reliability metric (inverse BER model)
Reliability = zeros(1, length(SNR));
for i = 1:length(SNR)
    % simple channel model effect
    H = (randn(Nr, Nt) + 1j*randn(Nr, Nt)) / sqrt(2);
    noise_power = 1 / SNR(i)
    % reliability increases with SNR
    Reliability(i) = 1 - exp(-SNR(i)/10);
end
figure
% -------- Graph 1: SNR vs Reliability --------
subplot(2,1,1)
plot(SNR_dB, Reliability, '-o', 'LineWidth', 2)
title('MIMO Reliability vs SNR')
xlabel('SNR (dB)')
ylabel('Reliability (0 to 1)')
grid on
% -------- Graph 2: SNR vs Noise Effect --------
subplot(2,1,2)
plot(SNR_dB, noise_power * ones(size(SNR_dB)), '-s', 'LineWidth', 2)

title('Noise Power Variation with SNR')
xlabel('SNR (dB)')
ylabel('Noise Power')
grid on
```

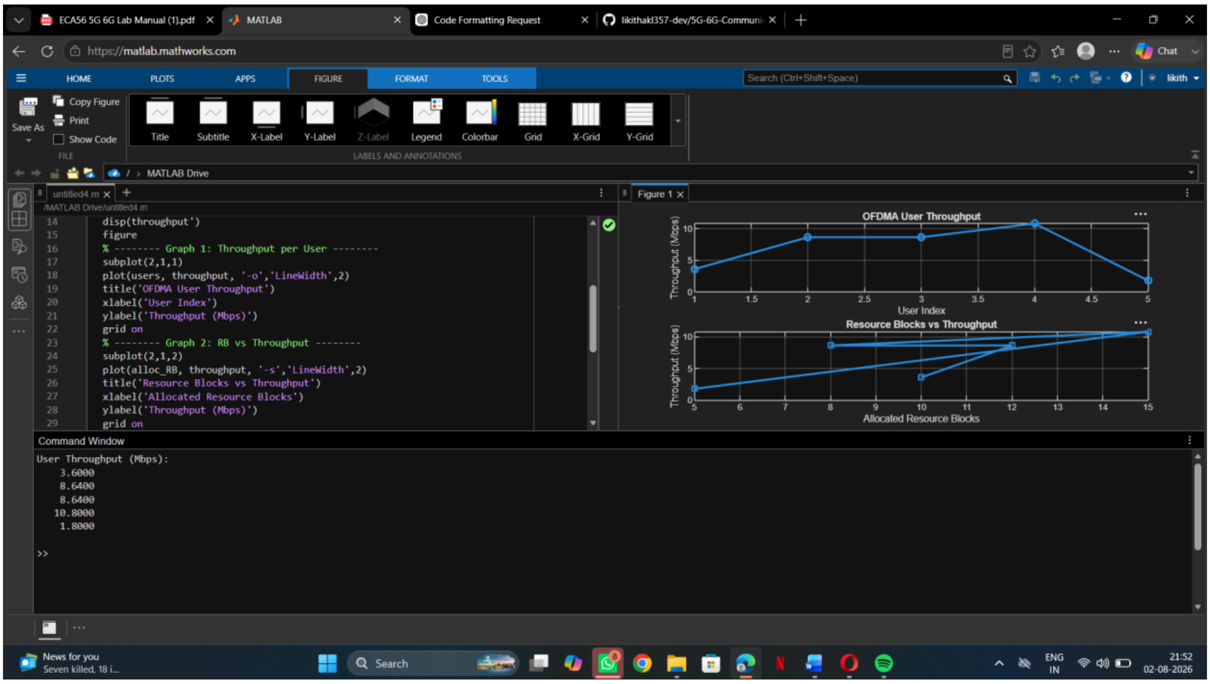


OBJECTIVE 2 CODE :

```
clc;
clear;
close all;
% Resource Block parameters
rb_bw = 180e3;   % 180 kHz per RB
% Users
users = 1:5;
% Allocated Resource Blocks per user
alloc_RB = [10 12 8 15 5];
% Modulation efficiency (bits/symbol)
mod_eff = [2 4 6 4 2];   % QPSK, 16QAM, 64QAM etc.
% Throughput calculation (Mbps)
throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;
% Display results
disp('User Throughput (Mbps):')
disp(throughput')
figure
% -------- Graph 1: Throughput per User --------
```

```
subplot(2,1,1)
plot(users, throughput, '-o', 'LineWidth', 2)
title('OFDMA User Throughput')
xlabel('User Index')
ylabel('Throughput (Mbps)')
grid on
% -------- Graph 2: RB vs Throughput --------
subplot(2,1,2)
plot(alloc_RB, throughput, '-s', 'LineWidth', 2)
title('Resource Blocks vs Throughput')
xlabel('Allocated Resource Blocks')
ylabel('Throughput (Mbps)')
grid on
```

RESULT :



Result:

1. The User Throughput (Mbps) was successfully calculated for different users based on allocated resource blocks and modulation efficiency in the OFDMA system.

2. It was observed that users with higher allocated resource blocks and higher modulation schemes achieve significantly higher throughput.

3. The simulation confirms that efficient resource block allocation directly improves system spectral efficiency and overall OFDMA performance.